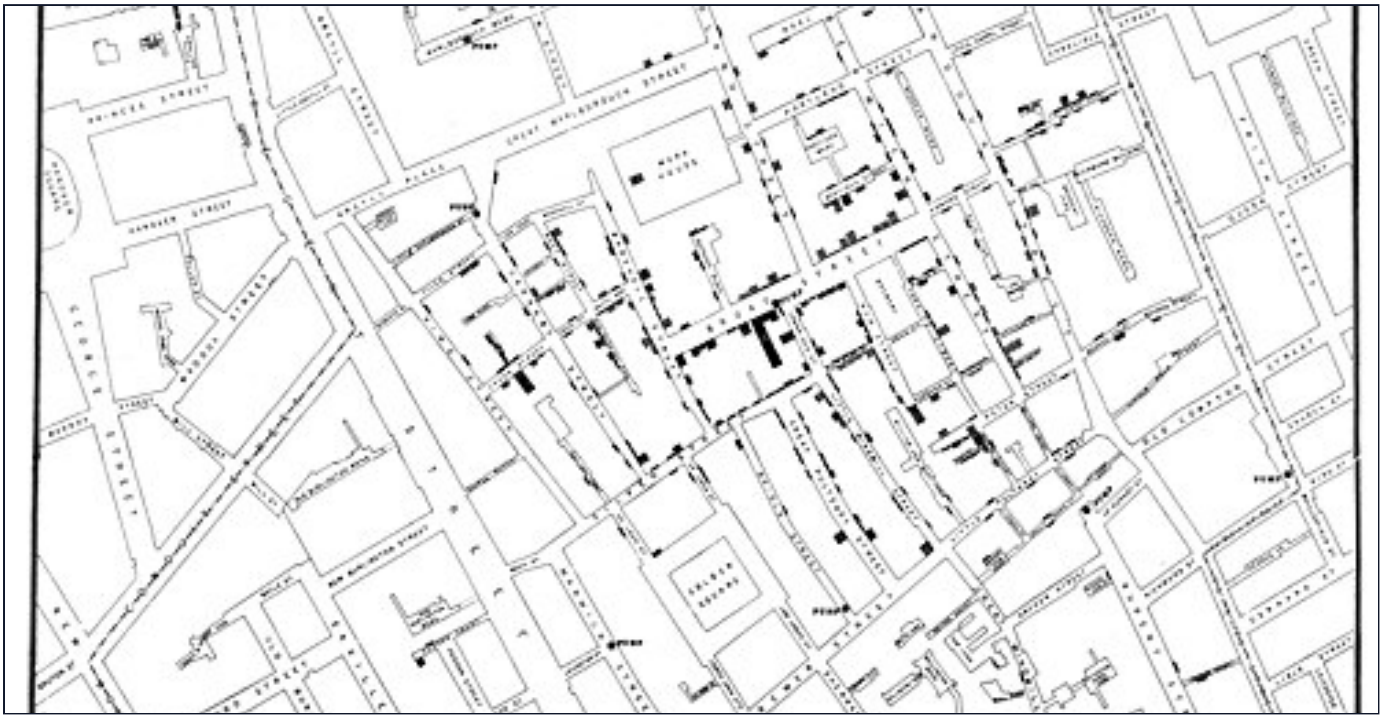
c. AD 43 – PRESENT

WATER & SANITATION THROUGH TIME

Engineering, Epidemics, and the Sanitary Revolution



Dr John Snow's Landmark 1854 Cholera Spot Map of Soho, London. By plotting fatalities around the Broad Street pump, Snow pioneered epidemiology and proved cholera was waterborne long before bacteria could be seen.

HISTORICAL OVERVIEW

Two Millennia of Public Health

For two thousand years, the survival of British communities has depended on their mastery of water and waste. From monumental Roman aqueducts and communal bathhouses to medieval conduit systems and the deadly horrors of the Black Death, societies struggled against pestilence with limited scientific understanding. The Industrial Revolution brought catastrophic crisis, packing millions of labourers into unventilated slums where sewage seeped into drinking wells. It took the terror of Asiatic cholera, the political panic of the 1858 Great Stink, and the pioneering genius of Dr John Snow, Sir Joseph Bazalgette, and Louis Pasteur to overturn ancient miasma beliefs and compel the Victorian state to build modern, life-saving infrastructure.

COURSE SPECIFICATION & DISCIPLINARY READING DIRECTORY

5 ENQUIRIES

Enquiry 01	**Roman Britain: Engineering & Public Health (AD 43–410)**	*"How much progress did the Romans make in public health?"*	**Page 3**
Enquiry 02	**The Middle Ages: Towns & Monasteries (1066–1500)**	*"Why did public health decline during the Middle Ages?"*	**Page 4**
Enquiry 03	**The Early Modern Era: Growth & Plague (1500–1700)**	*"To what extent did towns become filthier during the Early Modern period?"*	**Page 5**
Enquiry 04	**Industrial Squalor & The Great Stink (1750–1858)**	*"How did industrial squalor and the Great Stink force Britain to build modern sewers?"*	**Page 6**
Enquiry 05	**Germ Theory & The 1875 Revolution (1854–1900)**	*"How did Dr John Snow, Germ Theory, and the 1875 Act conquer waterborne disease?"*	**Page 7**

CORE CONCEPTS 🏛 Hydraulic Civil Engineering • 🔬 Miasma vs Germ Theory • ⚖ State Intervention vs Laissez-Faire • 👥 Public Duty vs Ratepayer Cost

KEY CHRONOLOGY & CAUSAL TURNING POINTS

Two Millennia of British Sanitation, Engineering Breakthroughs & Epidemics (c. AD 43–1875)

AD 75 — **Roman Public Baths and Aqueducts Constructed** [Ref 1.2]
Imperial engineers use gravity gradients to channel fresh mountain spring water across arched aqueducts into stone-walled cities like Londinium and Fishbourne Palace.

AD 410 — **Roman Legionary Withdrawal & Infrastructure Decay** [Ref 1.5]
With the collapse of Roman administrative authority and legions, aqueducts fall into disrepair, public bathhouses collapse, and Britons return to local wells and rivers.

c. 1165 — **Canterbury Cathedral Prior Wibert Piped Conduit System** [Ref 2.2]
Monks engineer sophisticated lead-piped spring conduits, settling tanks, and cloister lavatoriums, demonstrating monastic sanitary superiority over squalid towns.

1298 — **King Edward I Orders Clearance of Squalor in York** [Ref 2.3]
Appalled by human dung, animal entrails, and decomposing refuse corrupting the city air, King Edward I issues royal writs compelling bailiffs to pave and cleanse the streets.

1348 — **Arrival of the Black Death Pandemic at Melcombe Regis** [Ref 2.4]
The catastrophic bubonic and pneumonic plague strikes Dorset from Gascony, wiping out one-third to one-half of the English population amidst total medical ignorance.

1388 — **The First National Sanitary Statute (Cambridge Parliament)** [Ref 2.5]
Parliament passes Britain's first national public health law, imposing severe twenty-pound fines on butchers and tanners casting unwholesome dung and offal into public rivers.

1596 — **Sir John Harington Invents The Flushing Water Closet** [Ref 3.1]
Godson to Queen Elizabeth I patents the mechanical valve flushing toilet ("The Ajax"); rejected as a costly novelty without underground municipal sewer connections.

1613 — **Opening of the 38-Mile New River into London** [Ref 3.2]
Sir Hugh Myddelton finances an overland canal bringing clean Hertfordshire spring water into London reservoirs, supplying clean private water to subscribers.

1660 — **Samuel Pepys Records Cesspool Overflow Crisis** [Ref 3.3]
Diarist Samuel Pepys steps into his cellar to find his neighbour's overflowing cesspit has flooded it, illustrating ubiquitous early modern urban sanitation failures.

1665 — **The Great Plague of London & Miasma Terror** [Ref 3.4]
Over 100,000 Londoners die; medical authorities burn tar barrels and massacre stray animals believing poisonous atmospheric miasma causes the epidemic.

1831 — **Asiatic Cholera ("The Blue Death") Strikes Britain** [Ref 4.2]
Lethal waterborne epidemic arrives at the port of Sunderland, killing 52,000 Britons in months while doctors fruitlessly scrub streets with lime believing bad air caused disease.

1842 — **Edwin Chadwick Publishes The Landmark Sanitary Report** [Ref 4.3]
Poor Law commissioner Chadwick proves through statistics that slum filth causes disease and poverty, prompting the well-meaning but permissive 1848 Public Health Act.

1854 — **Dr John Snow Maps the Soho Broad Street Pump Outbreak** [Ref 5.2]
Dr John Snow plots 616 deaths on a cartographic spot map, removes the pump handle, and deduces that cholera is waterborne thirty years before bacteria can be identified.

June 1858 — **The Great Stink Forces Parliament to Finance Bazalgette's Sewers** [Ref 4.4]
A summer heatwave ferments 200,000 flushing toilets in the Thames, forcing MPs to flee Parliament and grant £3 million to build 82 miles of brick intercepting sewers.

Aug 1875 — **The Public Health Act 1875 Establishes the Compulsory State** [Ref 5.5]
Armed with Pasteur's Germ Theory and Disraeli's reforms, Parliament legally commands all local councils to provide clean water, drainage, and medical officers, ending laissez-faire.

ENQUIRY 01 *Was Roman water engineering built to protect ordinary people, or to show off imperial power and luxury?*

[1.1] Celtic Settlement & the Roman Conquest (AD 43)

How did the Roman invasion of AD 43 trigger an urban sanitation crisis?

Before the Roman conquest, Iron Age Britain was an agrarian society of small farming settlements and dispersed Celtic roundhouses. Waste disposal was simple and sustainable: families dug shallow cesspits in open fields that composted naturally into manure without poisoning freshwater streams or shallow wells.

Everything changed in AD 43 when Emperor Claudius invaded with four imperial legions. The Romans established cramped walled cities like Londinium (London) and Eboracum (York), packing thousands of soldiers and civilians together. Without municipal water engineering, urban populations faced catastrophic collapse from waterborne epidemics.

[1.2] Gravity Hydraulics: Aqueducts & Water Towers

How did Roman engineers use gravity to transport millions of gallons of water?

Roman engineers solved urban water shortages through mastery of gravity hydraulics. Because local rivers became fouled with urban filth, engineers surveyed distant mountain springs and hills to capture pure upland water supplies for growing provincial cities.

Engineers constructed massive stone aqueducts built on a precise mathematical downward gradient, channeling millions of gallons into castella water towers. From these storage reservoirs, lead and terracotta pipes distributed pressurized water across towns, prioritizing public street fountains, communal thermae baths, and wealthy private villas.

[1.3] The Thermae: Public Baths & Hypocaust Heating

How did monumental public bathhouses promote cleanliness and imperial culture?

The centerpiece of Roman urban hygiene was the communal public bathhouse (thermae), open to all citizens for a tiny copper coin. Bathers coated their skin in olive oil & bronze strigils were used by enslaved attendants to scrape away accumulated sweat and grime.

Bathers moved between heated steam chambers and the invigorating frigidarium cold plunge pool to close their pores. An ingenious subterranean hypocaust heating system circulated furnace heat beneath suspended mosaic floors, making public baths essential civic hubs where Romans exercised, socialised, and washed.

[1.4] Public Latrines, the Tersorium & the Lead Paradox

Why did forensic science expose severe flaws in Roman hygiene and health?

Roman towns featured communal stone latrines flushed continuously by bathhouse wastewater, where citizens sat together without privacy cubicles. For cleaning, Romans shared a communal sponge on a stick (tersorium), rinsed in a running saltwater floor channel between users.

Despite monumental engineering, archaeological excavations reveal widespread contamination. Warm bath water was rarely changed, exposing bathers to microscopic intestinal parasite eggs that spread whipworm and dysentery. Furthermore, drinking water delivered through toxic lead water pipes caused chronic plumbism poisoning among urban populations.

[1.5] Historical Verdict: Romanitas, Imperial Prestige & The AD 410 Collapse

Why did Roman water engineering abruptly collapse after AD 410?

Modern historians judge Roman sanitation as a reflection of imperial power & vanity rather than modern medical understanding. Roman governors constructed monumental aqueducts and marble baths primarily to project Romanitas and civilised prestige across newly conquered barbarian provinces.

The fatal flaw of Roman hygiene was its total dependence on centralized imperial administration. When vanishing Roman legions withdrew from Britain in AD 410, complex maintenance collapsed, leaving a thousand-year sanitation void as urban aqueducts decayed into overgrown rubble.

ENQUIRY 02 *Was medieval public health squalor caused by a lack of money, or by mistaken medical beliefs like miasma?*

[2.1] The Medieval Borough Boom & Overhanging Jetties

Why did the growth of medieval towns overwhelm traditional water supplies?

After the Roman legions abandoned Britain in AD 410, collapsed Roman stone aqueducts and centralized infrastructure fell into ruin. As the Anglo-Saxons and Normans rebuilt society, population centers shifted toward river valleys, river crossings, and fortified boroughs. By the twelfth century, booming market boroughs emerged across England, with thousands of traders crowding into unpaved muddy thoroughfares and narrow lanes.

Unlike Roman stone conduits, early medieval towns lacked centralized drainage. Butchers, tanners, and dyers operated directly in the streets: butchers slaughtered livestock on public pathways, throwing offal into gutters; tanners soaked animal hides in vats of urine; and dyers emptied toxic chemical dyes directly into rivers that downstream families used for drinking water.

[2.2] Municipal Waste Battles: Rakers, Gong Farmers & Court Leet Fines

How did medieval towns attempt to manage human and animal waste?

In medieval towns like London, York, and Coventry, municipal corporations fought a constant battle to manage waste. Narrow streets were lined with overcrowded timber tenements, where families constructed wooden privies over private backyard cesspits.

To prevent overflows, householders paid night-time gong farmers to shovel human excrement into barrels for rural crop fertiliser. Town bailiffs appointed municipal rakers to clear street rubbish, while Court Leet fines & rakers punished butchers and tanners who dumped offal into public thoroughfares.

[2.3] Monastic Hydraulics: Canterbury Priory's Five Settling Tanks (c.1165)

Why did medieval monasteries enjoy far cleaner water than crowded towns?

The sharpest contrast to crowded town squalor existed inside medieval religious houses. Monasteries engineered superior water infrastructure, with settling tanks & lead conduits carrying pure spring water from distant hills directly into monastic kitchens, infirmaries, and latrines.

Monks lived in segregated monastic cloisters shielded from outside contagion, designing multi-stage filtration where water flushed latrines built over running rivers. Establishments like Titchfield Abbey fishponds and Canterbury Priory managed advanced fisheries and clean running conduits, combining immense monastic wealth with technical literacy.

[2.4] The Black Death Catastrophe: Bubonic Plague & Miasma Dogma (1348)

How did the 1348 Black Death challenge medieval understanding of disease?

In the summer of 1348, an unprecedented catastrophe struck Britain: the **Black Death** (bubonic plague). Carried by the bacterium *Yersinia pestis* living in fleas on black rats, the plague swept along trade routes, killing between thirty and fifty percent of Britain's population in under eighteen months. Medieval medical knowledge could not explain the contagion: physicians blamed planetary alignments, divine wrath, and poisonous foul air (miasma).

In response to plague, authorities attempted public health interventions. Town councils barred travellers from infected regions and enacted forty-day port quarantines (*quarantena*). In London, cemetery space ran out within weeks, forcing the construction of emergency mass burial pits at Smithfield and East Smithfield where corpses were laid in tiers.

[2.5] Historical Verdict: The 1388 Statute of Cambridge & Peasant Realities

Did public health genuinely collapse, or have Victorians exaggerated medieval squalor?

Modern historians like Carole Rawcliffe have pioneered Carole Rawcliffe revisionism, demonstrating that medieval civic leaders actively fought urban filth. In 1388, Parliament passed the landmark Statute of Cambridge (1388), introducing compulsory river clean-up laws that levied massive £20 fines for dumping animal entrails into waterways.

In Coventry, town officials banned waste dumping in the River Sherbourne and constructed designated municipal waste dumps outside town walls. Medieval sanitation struggled not from civic apathy, but because town bailiffs lacked police power to enforce regulations across rapidly growing populations.

ENQUIRY 03 *Was early modern filth caused by population explosion, or the lack of running water to make inventions work?*

[3.1] The Great Metropolis: Tudor London's Population Boom & Overhangs

How did London's rapid expansion create an unprecedented sanitation crisis?

Between 1500 and 1700, England underwent a dramatic demographic transformation as national population doubles from roughly three million to over five million. By 1665, London was home to over 500,000 London residents crammed into towering, cantilevered timber tenements that jutted out over narrow streets, blocking sunlight and trapping sewer stench at ground level.

While wealthy merchants lived in spacious brick mansions in newly developed suburbs like Covent Garden, poor working families crowded into subdivided rooms inside damp alleys. Rapid urban migration completely overwhelmed traditional Tudor waste systems, creating severe sanitation crises across the capital.

[3.2] The Royal Novelty: Sir John Harington's Flush Toilet (1596)

Why did Sir John Harington's revolutionary flush toilet fail to clean up Britain?

In 1596, Queen Elizabeth's godson, Sir John Harington, invented the world's first flushing water closet, known as the 'Ajax'. Harington installed a working model at Richmond Palace, designing an ingenious valve-operated cistern that released water to wash waste into a basement vault.

However, the invention failed to transform Tudor society. The total absence of running water mains and town sewer networks meant the flush toilet remained an isolated royal curiosity. Ordinary Elizabethans continued relying on pottery chamber pots and wooden close-stools for another two centuries.

[3.3] Private Water for the Wealthy: Sir Hugh Myddelton & The New River (1613)

How did private enterprise attempt to bring clean spring water into London?

As London's medieval parish wells became dangerously polluted by expanding cesspits, wealthy commercial entrepreneurs stepped forward. In 1609, goldsmith and financier Sir Hugh Myddelton funded the construction of the New River to bring clean water to the expanding capital.

Myddelton engineered an ambitious 38-mile man-made canal, bringing fresh spring water from Hertfordshire to Islington. Water was distributed across London through a pioneering network of hollowed elm tree trunk pipes, providing reliable drinking water to wealthy subscribers and civic conduit fountains.

[3.4] Nightsoil and Leaking Cesspits: The Misery of Samuel Pepys' Cellar (1660)

What does Samuel Pepys' diary reveal about domestic sanitation in Stuart London?

The grim reality of seventeenth-century domestic sanitation is vividly recorded in Samuel Pepys's famous diary. In October 1660, Pepys suffered the horrific ordeal of stepping into neighbour's waste when entering his basement, discovering a neighbour's overflowing privy had flooded into his cellar.

Because urban cellars were poorly constructed, fractured brick cesspits routinely cracked and leaked raw sewage into adjoining properties and shallow drinking wells. Householders had to pay nightsoil men & filth carts to shovel out subterranean cesspools at midnight and cart foul waste beyond city walls.

[3.5] Historical Verdict: The Great Plague of 1665 & The Limits of Science

Why did medical responses to the 1665 Great Plague remain fundamentally medieval?

The Early Modern era presents a striking paradox. While the Scientific Revolution made dramatic advances in physics and astronomy, medical theory remained trapped in medieval beliefs. When the **Great Plague of 1665** struck London, causing over 100,000 plague fatalities in seven months, authorities still blamed miasma and planetary alignment.

The Lord Mayor ordered the slaughter of over 40,000 dogs and 200,000 cats (the very predators that hunted plague rats), while watchmen locked infected families inside boarded-up houses marked with red crosses. Citizens protected themselves by pounding tobacco & pomanders filled with perfumed herbs, completely unaware that rat fleas were transmitting the disease.

ENQUIRY 04 *Was Victorian cholera conquered by civil engineering and fear of the Great Stink, or did politicians act out of genuine care for the poor?* 1750–1858

[4.1] 1780–1840 Boomtown Squalor: Back-to-Backs, Cellar Dwellings & Privy Middens

How did rapid industrial urbanisation create unprecedented slum squalor in British factory boomtowns?

Between 1780 and 1850, the **Industrial Revolution** transformed Britain from an agrarian society into the world's first industrial economy. Millions of rural workers migrated to rapidly expanding steam-powered factory boomtowns like Manchester, Birmingham, and Leeds. In these industrial hubs, speculative builders squeezed thousands of unventilated back-to-back slums into tight courtyards, lacking running water, ventilation, or drains.

Entire streets of sixty people shared a single privy over shared overflowing privy middens, which spilled human waste across unpaved courts where children played. Life expectancy in Manchester slums plummeted to just nineteen years for working-class labourers, compared to forty years in rural agricultural villages.

[4.2] 1831–1832 The Blue Death: Asiatic Cholera Strikes & The Miasma Trap

Why did the arrival of Asiatic cholera in 1831 terrorize Victorian Britain and expose the failure of miasma theory?

In October 1831, a terrifying new disease arrived at the Sunderland Baltic shipping port from the Baltic: **Asiatic cholera**. Caused by a waterborne bacterium, cholera killed rapidly through catastrophic dehydration, turning victims' skin blue as blood thickened. During the 1831–32 epidemic, cholera claimed over 52,000 dehydration fatalities across Great Britain.

Because medical doctors still clung to the tar barrels & miasma fallacy, town councils burned barrels of pitch and tar in streets to purify foul air, completely failing to realise that human sewage leaking into parish water pumps was spreading the deadly infection.

[4.3] 1842–1848 Edwin Chadwick's Sanitary Report & The Permissive 1848 Act

Why did Edwin Chadwick's 1842 Report fail to clean up Britain despite proving that filth caused disease?

The first systematic breakthrough in public health came not from a doctor, but from civil servant **Edwin Chadwick**. In 1842, Chadwick published his groundbreaking *Report on the Sanitary Condition of the Labouring Population*. Using statistical data, Chadwick proved that epidemic disease and poverty were directly caused by slum filth, defective drainage, and contaminated water supplies.

Chadwick argued that cleaning up towns would save taxpayer money by reducing workhouse dependency. His report pressured Parliament to pass the **Public Health Act of 1848**, which established the first General Board of Health. However, because the 1848 Act was strictly *permissive* rather than compulsory, laissez-faire town councils refused to spend money on expensive sewer systems.

[4.4] June 1858 The Toxic Tide & The Great Stink of London

How did the environmental catastrophe of the 'Great Stink' finally shatter Victorian laissez-faire inertia?

By 1855, London was the commercial capital of the British Empire, home to nearly three million people. The widespread adoption of over 200,000 flushing water closets created an environmental disaster: water closets flushed massive volumes of raw human waste directly into old storm drains leading into the River Thames, turning the city's primary drinking source into an open sewer.

In the scorching summer of 1858, temperatures exceeded 35°C, fermenting Thames sludge into an overpowering stench known as the **Great Stink**. Parliament panic erupted as MPs soaked parliamentary curtains in chloride of lime to neutralise the smell, hurriedly passing emergency legislation and voting a £3m grant to finance a complete overhaul of London's drainage network.

[4.5] 1858–1865 Sir Joseph Bazalgette's Civil Engineering Triumph: 82 Miles of Sewers

Why was Sir Joseph Bazalgette's intercepting sewer network considered the greatest engineering triumph of the Victorian age?

To save London, the Metropolitan Board of Works appointed brilliant civil engineer **Joseph Bazalgette**. Between 1859 and 1875, Bazalgette oversaw one of the greatest civil engineering feats in world history. He constructed **82 miles of underground intercepting sewers** along the north and south banks of the Thames, using 318 million bricks and newly invented Portland cement.

Bazalgette's sewers were engineered on a precise gravity gradient to carry wastewater eastwards, aided by massive steam pumping stations like Abbey Mills, Crossness, and Portsmouth's gravity gradient & Eastney pumping station, discharging sewage into the tidal Thames far downstream from central London.

ENQUIRY 05 *Was cholera defeated by scientific observation, laboratory discovery of bacteria, or the introduction of compulsory government legislation?* 1854–1900

[5.1] 1854 The Soho Cholera Outbreak & Miasma Dogma (1854)

Why did the medical establishment refuse to consider water as the cause of the 1854 Soho outbreak?

In late August 1854, a sudden cholera outbreak struck the crowded Soho Broad Street epicentre in central London. Within 250 yards of the intersection between Broad Street and Cambridge Street, over 500 deaths in ten days occurred, terrorising local families and causing three-quarters of the neighbourhood to flee.

At the time, the medical establishment was gripped by entrenched miasma medical dogma, insisting that the summer heat had caused decaying organic matter in Soho's cesspools to release poisonous miasmatic vapours into the air, suffocating residents while they slept.

[5.2] 1854 The Medical Detective: Dr John Snow & The Broad Street Investigation

How did Dr John Snow use cartographic mapping and anomalous data to prove cholera was waterborne?

One London doctor rejected the miasma consensus: **Dr John Snow**, a leading physician and anaesthetist who had administered chloroform to Queen Victoria during childbirth. Snow reasoned that because cholera attacked the digestive tract, the infective agent had to be ingested in contaminated water rather than inhaled into the lungs. Snow conducted an empirical door-to-door epidemiological investigation, plotting an empirical cholera spot map marking every fatality.

The visual pattern revealed dense Broad Street pump clustering where cases surrounded the pump. Crucially, the nearby Lion Brewery had zero fatalities because its 70 workers drank free beer instead of water, while the local workhouse had its own private well.

[5.3] 1854–1855 The Cesspool Revelation: 40 Broad Street & The Removed Handle

What physical evidence finally proved how the Broad Street pump well had become contaminated?

On the evening of Thursday 7 September 1854, Dr John Snow presented his findings to the parish authorities, and the next morning officials removed Broad Street pump handle. The epidemic collapsed almost immediately.

Subsequent investigations by Reverend Henry Whitehead revealed the source of contamination: at 40 Broad Street, infant Frances Lewis had contracted cholera in late August. Her mother had washed soiled nappies into an old cesspool less than three feet from the public well, where fractured brickwork & leaking cesspool contents percolated directly into the drinking water drawn by hundreds of residents.

[5.4] 1861–1884 Louis Pasteur's Germ Theory (1861) & Robert Koch's Cholera Bacterium (1884)

Why was laboratory bacteriological proof necessary to finally overthrow the ancient miasma theory?

Despite Snow's proof, miasma dogma lingered for years. The definitive scientific breakthrough came from French chemist **Louis Pasteur**, who published his **Germ Theory of Disease** in 1861. Through his famous swan-neck flask experiments, Pasteur proved that microorganisms in the air caused fermentation and decay, disproving spontaneous generation.

Two decades later, in 1884, German microbiologist **Robert Koch** isolates Vibrio cholerae under the microscope, providing indisputable bacteriological proof that confirmed Dr John Snow's waterborne theory and destroyed miasma theory forever.

[5.5] 1875 The Public Health Act of 1875 & The Fall of Laissez-Faire

Why did the 1875 Public Health Act mark the final death of laissez-faire in Britain?

The convergence of scientific certainty, Bazalgette's engineering triumphs, and working-class voting power led directly to the **Public Health Act of 1875**. Unlike earlier legislation, the 1875 Act was strictly **compulsory** across Great Britain.

Town councils were legally required to appoint compulsory municipal officers & water inspectors, Medical Officers of Health, supply clean piped water, collect refuse, and construct covered sewers. The 1875 Act permanently ended laissez-faire ideology in sanitation, establishing clean water and public health as universal legal rights for all British citizens.

MASTER KNOWLEDGE RETRIEVAL QUIZ

25 Essential Disciplinary Recall Questions across 5 Canonical Enquiries (AD 43–1900)

Partner Quiz or Solo Practice: Use these 25 essential questions to test your chronological schema. All questions cross-reference specific textbook paragraph ranges (e.g. Ref [4.1]–[4.5]). Strictly zero answers are printed—retrieve facts purely from memory!

TOPIC CARD 1: ROMAN BRITAIN (AD 43–410)

p. 3 • Ref [1.1]–[1.5]

Q1. What architectural structure transported millions of gallons of fresh spring water across valleys into Roman cities?
Q2. How did Roman engineers heat public bathhouses like the Thermae?
Q3. What metal was used to make Roman water supply pipes, later linked to chronic poisoning?
Q4. What was the communal sponge on a stick used in Roman public latrines called?
Q5. What happened to Britain's public health infrastructure after the Roman legions withdrew in AD 410?

TOPIC CARD 2: THE MIDDLE AGES (1066–1500)

p. 4 • Ref [2.1]–[2.5]

Q6. Why did medieval monasteries enjoy significantly better water supplies than medieval towns?
Q7. What was the name of the 12th-century hydraulic waterworks plan preserved at Canterbury Cathedral?
Q8. What trade wastes from butchers and tanners severely contaminated medieval rivers like the Fleet?
Q9. In what year did the Black Death arrive in England through the port of Melcombe Regis in Dorset?
Q10. What was the dominant medical theory of disease throughout the Middle Ages?

TOPIC CARD 3: THE EARLY MODERN ERA (1500–1700)

p. 5 • Ref [3.1]–[3.5]

Q11. Who invented the world's first mechanical flushing water closet in 1596, named "The Ajax"?
Q12. What 38-mile canal brought fresh spring water from Hertfordshire into London reservoirs in 1613?
Q13. What happened in Samuel Pepys' cellar on 20 October 1660 due to poorly constructed cesspits?
Q14. What epidemic struck London in 1665, killing nearly 100,000 people in eighteen months?
Q15. What symbol was painted on the doors of quarantined houses during the 1665 Great Plague?

TOPIC CARD 4: INDUSTRIAL GROWTH & THE GREAT STINK (1750–1858)

p. 6 • Ref [4.1]–[4.5]

Q16. What type of unventilated, high-density housing was built for factory workers in boomtowns like Manchester?
Q17. In what year and northern port town did Asiatic cholera ("The Blue Death") first strike Great Britain?
Q18. Who published the landmark 1842 Report proving that slum filth caused premature disease and economic poverty?
Q19. What environmental crisis occurred in June 1858 when hot weather fermented raw sewage in the Thames?
Q20. How many miles of underground intercepting brick sewers did Sir Joseph Bazalgette construct across London?

TOPIC CARD 5: GERM THEORY & THE 1875 REVOLUTION (1854–1900)

p. 7 • Ref [5.1]–[5.5]

Q21. Where did the catastrophic 1854 cholera outbreak investigated by Dr John Snow take place?
Q22. What investigation tool did Dr John Snow create to map the cluster of deaths around the Broad Street pump?
Q23. Why did none of the 70 workers at the Lion Brewery on Broad Street contract cholera in 1854?
Q24. Who published the Germ Theory of Disease in 1861, proving microscopic organisms caused decay and infection?
Q25. What made the Public Health Act of 1875 fundamentally more effective than the earlier 1848 Act?